

one of the key levers to shorten the specification-to-production timeline for the DTV-media chip and thus achieve faster time to market.

Keeping pace with next-generation technologies

From the application software standpoint, OEMs need to adopt next-generation technologies like green TV and personal home assist, integrate new-age operating systems like Android and Google TV, handle integration issues of middleware stacks of emerging and existing standards, and integrate/manage the complete smart TV ecosystem for content. Ensuring the integration of DTV SoC with next-generation technologies is a challenge for DTV OEMs. Let us analyse the challenge posed by next-generation green TV and personal home assist technologies.

Green TV. Today, upgrades and eWaste are seen as a pressing issue in the DTV landscape. On an average, a consumer in the US or Japan is willing to consider buying a new DTV every

four years. With technology changing at the speed of light, the customer might soon find his DTV outdated. This is driving the demand for a future-proof and truly green TV.

With the help of over-the-Internet software upgrades for feature enhancements and a modular plug-and-play PCB architecture, it is possible to architect a future-proof DTV. The modular plug-and-play PCB architecture will have a fixed motherboard carrying display and power electronics and a pluggable Media-SoC board hosting key DTV engines.

Personal home assist. Similarly, there is an increasing demand for 'home assistant' feature among DTV consumers wanting an experience superior to 'personal assistant' available on today's ultra-high-end smartphones. Such a unified home assistant would mean a sophisticated and self-learning artificial intelligence based voice cognition application running on the DTV OS, tightly linked at the system software level to native applications supporting

valuable use cases at home. This will look like a keyboard-less virtual home assistant on the DTV that will accept authenticated commands from the user to decode and control devices at home over the home network and provide valuable status messages and calendar items on the screen.

The applications of this virtual home control center will be endless. From a hardware context, this will mean freeing up one or more dedicated MCU cores for running sophisticated home-assist software, and stepping up the chip performance.

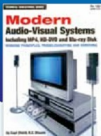
Road ahead

The digital TV landscape will get more complex with new entrants like IP vendors, new semiconductor and OEMs entering and growing dynamism of the consumer-driven product lifecycle. This will result in increasing collaboration in the DTV landscape. ●

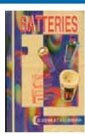
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